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DSC623 Database Project - Implementation
Case Study - Reliable Rentals

Introduction

This report presents the implementation of the Reliable Rentals database using Python and SQLite. The implementation is based on the conceptual model developed in part 1 and the logical model developed in part 2.

The purpose of this part of the project is to create the complete database schema, apply the integrity constraints identified in the logical design, populate each relation with sample data, and implement the five user transactions using embedded SQL. This report documents the implemented relations, constraints, sample data, and query results for the Reliable Rentals case study.

Connection to Previous Project Parts

Part 1 focused on identifying the main entities, relationships, multiplicity constraints, attributes, and key attributes for the Reliable Rentals system. Part 2 then translated that conceptual design into a logical data model by deriving relations, identifying foreign keys, validating the design through normalization to 3NF, validating the model against 5 user transactions, and defining integrity constraints.

Part 3 is the physical implementation of that design. The database schema created in Python and SQLite reflects the final logical model from Part 2, and the 5 embedded SQL queries implemented in Python correspond to the 5 user transactions previously defined.

Assumptions Used in the Design

The implementation follows the assumptions stated in Part 2 of the project

1. Each model belongs to exactly one make.
2. A make can have many models
3. All vehicles of the same model share the same engine size and capacity
4. Each vehicle has a unique registration number regardless of model or make

These assumptions support the separation of the Models relation from the Vehicle relation and justify the use of registrationNumber as the unique identifier for each vehicle. They also support the structure of the logical model that was implemented in this phase.

Database Creation and Implementation

The database is implemented in the Python file schema_and_data.py. The script first removes any previous version of the database file and creates a fresh SQLite database named vehicle_hire.db.

After creating the connection, the script enables foreign key enforcement. This supports referential integrity between related relations in the database.

The script then creates the six tables using SQL CREATE TABLE statements embedded in Python. The resulting schema reflects the relations identified in Part 2 and includes key constraints and additional validation constraints where appropriate.

Description of Implemented Relations

Site
The Site relation stores information about the company's rental outlets. It contains siteNumber, phoneNumber, faxNumber, and address. The primary key is siteNumber.

Models
The Models relation stores model and make information for vehicles. It contains model and make, where model is the primary key.

Client
The Client relation stores customer information. It contains clientNumber, name, address, phoneNumber, DOB, and licenseNumber. The primary key is clientNumber.

Staff
The Staff relation stores employee information. It contains staffNumber, name, homeAddress, phoneNumber, DOB, sex, startDate, jobTitle, salary, and siteNumber. The primary key is staffNumber, and siteNumber is a foreign key referencing Site(siteNumber).

Vehicle
The Vehicle relation stores information about all vehicles available for hire. It contains registrationNumber, model, engineSize, capacity, currentMileage, dailyHireRate, curLocation, and siteNumber. The primary key is registrationNumber. The attributes model and siteNumber function as foreign keys referencing Models(model) and Site(siteNumber) respectively.

HireAgreement
The HireAgreement relation stores rental transaction data. It contains hireNumber, clientNumber, hireStartDate, hireTerminationDate, vehicleRegistrationNumber, startMileage, and endMileage.

The primary key is hireNumber, while clientNumber and vehicleRegistrationNumber function as foreign keys referencing Client(clientNumber) and Vehicle(registrationNumber).

Sample Data Population

After creating the schema, the schema_and_data.py script inserts sample data into every relation. Each relation contains at least 5 tuples, satisfying the project requirement.

The Database Contains:

- 5 Site records
- 5 Model records
- 5 Client records
- 5 Staff records
- 5 Vehicle records
- 5 Hire Agreement records

Verification of Table Contents

The transactions.py file begins by displaying the contents of all six relations using SELECT * queries. These verification queries confirm that the database was created correctly and that the sample data was inserted successfully.

```
Site (initial)
 siteNumber  phoneNumber    faxNumber                          address
          1 305-555-0101 305-555-0102           100 Ocean Dr, Miami, FL
          2 305-555-0201 305-555-0202       500 Biscayne Blvd, Miami, FL
          3 407-555-0301 407-555-0302        200 Orange Ave, Orlando, FL
          4 813-555-0401 813-555-0402       300 Bayshore Blvd, Tampa, FL
          5 904-555-0501 904-555-0502 400 Riverside Dr, Jacksonville, FL
```

Models (initial)

model	make
Civic	Honda
Corolla	Toyota
F-150	Ford
Model 3	Tesla
Wrangler	Jeep

Client (initial)

clientNumber	name	address	phoneNumber	DOB	licenseNumber
101	Alice Nguyen	12 Palm St, Miami, FL	305-111-1001	1990-04-12	FL-DL-1001
102	Bob Martinez	34 Coral Way, Miami, FL	305-111-1002	1985-08-23	FL-DL-1002
103	Carla Singh	78 Sunset Ave, Orlando, FL	407-111-1003	1992-11-05	FL-DL-1003
104	Diego Alvarez	5 Bay St, Tampa, FL	813-111-1004	1978-02-19	FL-DL-1004
105	Evelyn Park	90 River Rd, Jacksonville, FL	904-111-1005	1995-07-30	FL-DL-1005

Staff (initial)

staffNumber	name	homeAddress	phoneNumber	DOB	sex	startDate	jobTitle	salary	siteNumber
1	Sebastian Cruz	1 Beach Rd, Miami, FL	305-222-2001	1988-03-15	M	2018-06-01	Manager	75000	1
2	Maria Lopez	2 Palm Ln, Miami, FL	305-222-2002	1991-09-21	F	2020-01-10	Sales Agent	48000	1
3	James Chen	3 Coral Ct, Miami, FL	305-222-2003	1986-12-04	M	2017-03-22	Mechanic	52000	2
4	Priya Patel	4 Lake Dr, Orlando, FL	407-222-2004	1993-05-17	F	2021-08-15	Sales Agent	47000	3
5	Tom Reilly	5 Bay Ave, Tampa, FL	813-222-2005	1980-10-30	M	2015-11-02	Manager	78000	4

Vehicle (initial)

registrationNumber	model	engineSize	capacity	currentMileage	dailyHireRate	curLocation	siteNumber
FL-AAA-001	Civic	1.8	5	22000	45	Miami Lot A	1
FL-AAA-002	Corolla	1.8	5	18500	42	Miami Lot A	1
FL-BBB-003	F-150	5.0	3	35000	80	Miami Lot B	2
FL-CCC-004	Model 3	0.0	5	9000	95	Orlando Garage	3
FL-DDD-005	Wrangler	3.6	4	41000	75	Tampa Yard	4

HireAgreement (initial)

hireNumber	clientNumber	hireStartDate	hireTerminationDate	vehicleRegistrationNumber	startMileage	endMileage
1001	101	2026-04-20	None	FL-AAA-001	22000	NaN
1002	103	2026-04-15	None	FL-CCC-004	9000	NaN
1003	102	2026-03-01	2026-03-10	FL-BBB-003	34000	35000.0
1004	104	2026-02-14	2026-02-20	FL-DDD-005	40500	41000.0
1005	105	2026-01-05	2026-01-12	FL-AAA-002	18000	18500.0

Implementations of the Five User Transactions

```
T1: Staff at the site of vehicle FL-AAA-001
 staffNumber           name    jobTitle  siteNumber
           1 Sebastian Cruz     Manager           1
           2    Maria Lopez Sales Agent           1


T2: Address of client who hired FL-AAA-001 on 2026-04-20
 clientNumber         name                address
          101 Alice Nguyen 12 Palm St, Miami, FL


T3: Vehicle count at Sebastian's site
     staffName  siteNumber  vehicleCount
Sebastian Cruz           1             2


T4: Make/model of active hires by client 101
 hireNumber vehicleRegistrationNumber  make model hireStartDate
       1001                FL-AAA-001 Honda Civic    2026-04-20


T5: Site phone for vehicles rented by client 102
 hireNumber vehicleRegistrationNumber  siteNumber sitePhoneNumber                 siteAddress
       1003                FL-BBB-003           2    305-555-0201 500 Biscayne Blvd, Miami, FL
```

These 5 embedded SQL queries demonstrate that the implemented schema supports relationship navigation, multi-table joins, filtering, and aggregation in a way that is consistent with the logical model.

Python Program Structure

The implementation uses two Python files:

- schema_and_data.py creates the database, defines the schema, applies constraints, and inserts the sample data
- transactions.py verifies the contents of the database and executes the five required embedded SQL queries

The transactions.py script uses a helper function to execute SQL statements and display the results in a readable tabular format.

Conclusion

The Reliable Rental database was successfully implemented in Python and SQLite based on the conceptual and logical models developed in the earlier project phases. The implementation

includes the full schema, identified keys and integrity constraints, sample data for every relation, and the five embedded SQL queries required for validation.

The final implementation demonstrates that the database can support the main business operations of Reliable Rentals, including site management, vehicle management, client records, staff assignments, and hire agreements. The verification queries and transaction outputs confirm that the database was created and populated successfully.

Repository: https://github.com/DanielW136/DSC623-Final-Project